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VIVO Question of DevOps
(M23BIC607B)

Department of CSE-IC

Program–1: Create Project in Local and Remote Repository using Git and GitBash

1. What is Git?

Answer: Git is a distributed version control system used to track changes in source code.

2. What is Git Bash?

Answer: Git Bash is a command-line interface that allows users to execute Git commands on Windows.

3. What is a repository?

Answer: A repository (repo) is a storage location where project files and version history are maintained.

4. Difference between local and remote repository?

Answer: Local repository exists on the user's machine, while a remote repository is hosted on platforms like GitHub.

5. Which command initializes a Git repository?

Answer: `git init`

6. Which command checks repository status?

Answer: `git status`

7. How do you add files to staging area?

Answer: `git add filename` or `git add .`

8. How do you commit changes?

Answer: `git commit -m "message"`

9. How do you connect local repository to remote repository?

Answer: `git remote add origin <repository_url>`

10. How do you push code to GitHub?

Answer: `git push origin main`

Program-2: Fork, Merge, Conflict and Rebase using GitHub

11. What is Fork?

Answer: Fork creates a copy of another user's repository in your GitHub account.

12. What is Merge?

Answer: Merge combines changes from one branch into another.

13. What is a Merge Conflict?

Answer: A conflict occurs when Git cannot automatically combine changes.

14. What is Rebase?

Answer: Rebase moves or combines commits onto another branch.

15. Difference between Merge and Rebase?

Answer: Merge preserves history; rebase creates a linear history.

16. Which command is used for merging?

Answer: `git merge branch_name`

17. Which command is used for rebasing?

Answer: `git rebase branch_name`

18. Why is fork used in open-source projects?

Answer: To contribute without direct access to the original repository.

19. What is a Pull Request?

Answer: A request to merge code changes into another repository.

20. How do you resolve merge conflicts?

Answer: Edit conflicting files manually and commit the resolved changes.

Program-3: Interface Repository with Jenkins

21. What is Jenkins?

Answer: Jenkins is an open-source automation server used for CI/CD.

22. What is Continuous Integration?

Answer: It is the practice of automatically building and testing code changes.

23. What is a Jenkins Job?

Answer: A task or project configured in Jenkins.

24. How does Jenkins connect to GitHub?

Answer: Through repository URL and credentials/webhooks.

25. What is a Build?

Answer: Process of compiling and packaging source code.

26. What is a Jenkins Workspace?

Answer: Directory where Jenkins stores project files.

27. What is a Trigger in Jenkins?

Answer: An event that starts a build automatically.

28. What is a Jenkins Pipeline?

Answer: A series of automated build, test, and deployment stages.

29. What is Jenkinsfile?

Answer: A file containing pipeline configuration as code.

30. Why use Jenkins?

Answer: To automate software development workflows.

Program-4: Scheduling using Continuous Integration Tool

31. What is Scheduling?

Answer: Running jobs automatically at predefined times.

32. What is Cron?

Answer: A scheduling utility used to automate tasks.

33. What is Cron Expression?

Answer: A string representing schedule timing.

34. Example of Cron syntax?

Answer: 0 9 * * * runs every day at 9 AM.

35. Why schedule builds?

Answer: To automate repetitive tasks.

36. What is Poll SCM?

Answer: Jenkins periodically checks repository changes.

37. What is Build Periodically?

Answer: Jenkins executes jobs based on schedule.

38. What is Event-based Trigger?

Answer: Build starts when a specific event occurs.

39. Advantage of Scheduled Builds?

Answer: Saves manual effort.

40. What is Nightly Build?

Answer: Automated build executed during night hours.

Program-5: Selenium WebDriverIO and Reporting Tools

41. What is Selenium?

Answer: A tool used for automating web browsers.

42. What is WebDriverIO?

Answer: A Node.js framework for browser automation.

43. What is Test Automation?

Answer: Executing test cases automatically using scripts.

44. What is a Locator?

Answer: Mechanism used to identify web elements.

45. What is Assertion?

Answer: Verification of expected results.

46. What reporting tools are commonly used?

Answer: Allure, Extent Reports, HTML Reports.

47. What is Test Suite?

Answer: Collection of related test cases.

48. What is Headless Browser?

Answer: Browser running without GUI.

49. What is Selenium Grid?

Answer: Tool for parallel test execution.

50. Why use reporting tools?

Answer: To visualize test execution results.

Program-6: XPath and Element Identification

51. What is XPath?

Answer: XML Path Language used to locate elements.

52. What is Absolute XPath?

Answer: XPath starting from root node.

53. What is Relative XPath?

Answer: XPath starting from any node.

54. Example of Relative XPath?

Answer: `//input[@id='username']`

55. Which XPath is preferred?

Answer: Relative XPath.

56. What is contains() in XPath?

Answer: Function used for partial matching.

57. What is text() in XPath?

Answer: Function used to locate element based on text.

58. What is XPath Axis?

Answer: Method to navigate elements.

59. Example of XPath Axis?

Answer: parent, child, following-sibling.

60. Advantage of XPath?

Answer: Flexible element identification.

Program-7: Maven Lifecycle and Jenkins Integration

61. What is Maven?

Answer: Build automation and project management tool.

62. What is POM.xml?

Answer: Project Object Model configuration file.

63. What is Dependency?

Answer: External library required by project.

64. What is Maven Repository?

Answer: Storage location for dependencies.

65. What is Maven Lifecycle?

Answer: Sequence of build phases.

66. What is Compile Phase?

Answer: Converts source code into bytecode.

67. What is Test Phase?

Answer: Executes test cases.

68. What is Package Phase?

Answer: Creates deployable artifact.

69. Command to clean project?

Answer: mvn clean

70. Why integrate Maven with Jenkins?

Answer: To automate project build process.

Program-8: Docker Commands and Integration

71. What is Docker?

Answer: Containerization platform for applications.

72. What is a Container?

Answer: Lightweight executable package containing application and dependencies.

73. What is Docker Image?

Answer: Blueprint for creating containers.

74. Command to list containers?

Answer: `docker ps`

75. Command to list images?

Answer: `docker images`

76. What is Docker Hub?

Answer: Online repository for Docker images.

77. Command to run container?

Answer: `docker run image_name`

78. What is Dockerfile?

Answer: Script containing image creation instructions.

79. Difference between VM and Container?

Answer: Containers share OS kernel; VMs have separate OS.

80. Why use Docker?

Answer: Consistent application deployment.

Program-9: Agile Backlogs, Sprints, Tasks and jQuery

81. What is Agile?

Answer: Iterative software development methodology.

82. What is Product Backlog?

Answer: Prioritized list of project requirements.

83. What is Sprint?

Answer: Fixed-duration development cycle.

84. What is Sprint Backlog?

Answer: Tasks selected for current sprint.

85. What is User Story?

Answer: Requirement written from user perspective.

86. What is Task?

Answer: Small unit of work.

87. What is jQuery?

Answer: JavaScript library for DOM manipulation.

88. How to include jQuery?

Answer: Using CDN or local file.

89. What is AJAX in jQuery?

Answer: Technique for asynchronous data retrieval.

90. Advantage of jQuery?

Answer: Simplifies JavaScript coding.

Program–10: DevOps Framework Integration

91. What is DevOps?

Answer: Combination of Development and Operations practices.

92. What are DevOps tools?

Answer: Git, Jenkins, Maven, Docker, Selenium, Kubernetes.

93. What is CI/CD?

Answer: Continuous Integration and Continuous Deployment.

94. What is Automation?

Answer: Performing tasks without manual intervention.

95. What is Deployment?

Answer: Releasing application to target environment.

96. What is Monitoring?

Answer: Tracking system performance and health.

97. What is Infrastructure as Code?

Answer: Managing infrastructure through code.

98. What is Kubernetes?

Answer: Container orchestration platform. Kubernetes

99. Why integrate multiple DevOps tools?

Answer: To automate the complete software delivery pipeline.

100. What is the main goal of DevOps?

Answer: Faster, reliable, and continuous software delivery.

